
AWS EKS – Elastic Kubernetes Service Guide for Solutions Architects

1. What is AWS EKS?

- Fully managed Kubernetes service.
 - Handles Kubernetes control plane management.
 - Provides high availability and scalability.
 - Integrates with AWS ecosystem.
-

2. Core Components of EKS

Component	Description
Control Plane	Managed Kubernetes masters, auto-scaled & patched by AWS.
Worker Nodes	EC2 instances or Fargate used to run your workloads.
Cluster	Kubernetes cluster consisting of control plane + worker nodes.
Node Groups	Group of worker nodes managed as a unit.
Add-ons	Core Kubernetes components like CoreDNS, VPC CNI, and kube-proxy.

3. Key Features

- Managed control plane with automatic patching and upgrades.

- Supports self-managed or managed node groups.
 - Integration with IAM for authentication ([aws-iam-authenticator](#)).
 - Supports Fargate for serverless Kubernetes pods.
 - Supports Kubernetes-native networking via AWS VPC CNI plugin.
 - Deep integration with AWS services like ALB Ingress Controller, CloudWatch, CloudTrail, X-Ray.
-

4. Architecture Overview

- Control Plane runs in **AWS-managed AZs** across multiple AZs.
 - Worker nodes run in your VPC subnets.
 - Communication between control plane and worker nodes over secure connections.
 - Load balancing via **Kubernetes Services** and AWS ALB/NLB.
-

5. Creating an EKS Cluster

Using AWS Console:

1. Open EKS service.
2. Click **Create cluster**.
3. Enter:
 - Cluster name
 - Kubernetes version
 - Role (IAM role with EKS permissions)

- VPC and subnet selections (multi-AZ recommended)
4. Create the cluster.

Using AWS CLI:

```
aws eks create-cluster --name my-cluster --region us-west-2 \
--kubernetes-version 1.27 \
--role-arn arn:aws:iam::ACCOUNT_ID:role/EKS-ClusterRole \
--resources-vpc-config subnetIds=subnet-abc123,subnet-def456,securityGroupIds=sg-abc123
```

6. Worker Nodes Setup

- Create **Managed Node Group** via Console or CLI (simpler management).
 - Or launch **Self-Managed Nodes** (your own EC2 instances with [amazon-eks-node](#) AMI).
 - Node groups auto-register with control plane.
-

7. Networking in EKS

- Uses **AWS VPC CNI plugin** for pod networking.
 - Pods get native VPC IP addresses.
 - Supports **Security Groups for Pods** with **Amazon EKS Kubernetes Network Policies**.
 - Load Balancers:
 - ALB Ingress Controller for HTTP/HTTPS ingress.
 - NLB for TCP/UDP.
-

8. Authentication & Authorization

- EKS integrates with **AWS IAM** for authentication.
 - Uses `aws-iam-authenticator` to authenticate users.
 - RBAC manages authorization inside Kubernetes.
 - IAM Roles for Service Accounts (IRSA) allow assigning IAM permissions at pod level.
-

9. Monitoring and Logging

- Use **CloudWatch Container Insights** for cluster and pod metrics.
 - Centralize logs via **Fluentd** or AWS FireLens to CloudWatch Logs or third-party.
 - Use **AWS X-Ray** for distributed tracing.
 - Use **Prometheus** and **Grafana** for Kubernetes-native monitoring.
-

10. Scaling

- **Cluster Autoscaler**: Automatically scale worker nodes.
 - **Horizontal Pod Autoscaler (HPA)**: Scale pods based on CPU/memory.
 - **Vertical Pod Autoscaler (VPA)**: Adjust pod resource requests dynamically.
 - Use **Fargate** to scale pods serverlessly.
-

11. CI/CD with EKS

- Build Docker images → Push to **ECR**.
 - Deploy using **kubectl**, **eksctl**, **Helm**, or GitOps tools (ArgoCD, Flux).
 - Integrate with AWS CodePipeline, Jenkins, GitHub Actions.
-

12. Security Best Practices

- Enable **Private Endpoint** for EKS API.
 - Use **IAM Roles for Service Accounts**.
 - Network policies for pod isolation.
 - Use **Secrets Manager** or **SSM Parameter Store** for secrets.
 - Regularly patch Kubernetes versions.
 - Enable encryption at rest for etcd.
 - Enable audit logging.
-

13. Common Commands

Update kubeconfig for cluster access

```
aws eks update-kubeconfig --name my-cluster --region us-west-2
```

Create nodegroup with eksctl

```
eksctl create nodegroup --cluster my-cluster --name my-nodegroup --node-type t3.medium  
--nodes 3
```

Apply Kubernetes manifests

```
kubectl apply -f deployment.yaml
```

14. Tools & Ecosystem

- **eksctl**: Simplified CLI for cluster and nodegroup management.
 - **kubect**l: Kubernetes CLI.
 - **Helm**: Kubernetes package manager.
 - **AWS Load Balancer Controller**: Manage ALB ingress.
 - **Kube-prometheus-stack**: Monitoring.
-